

Tugas Machine Learning

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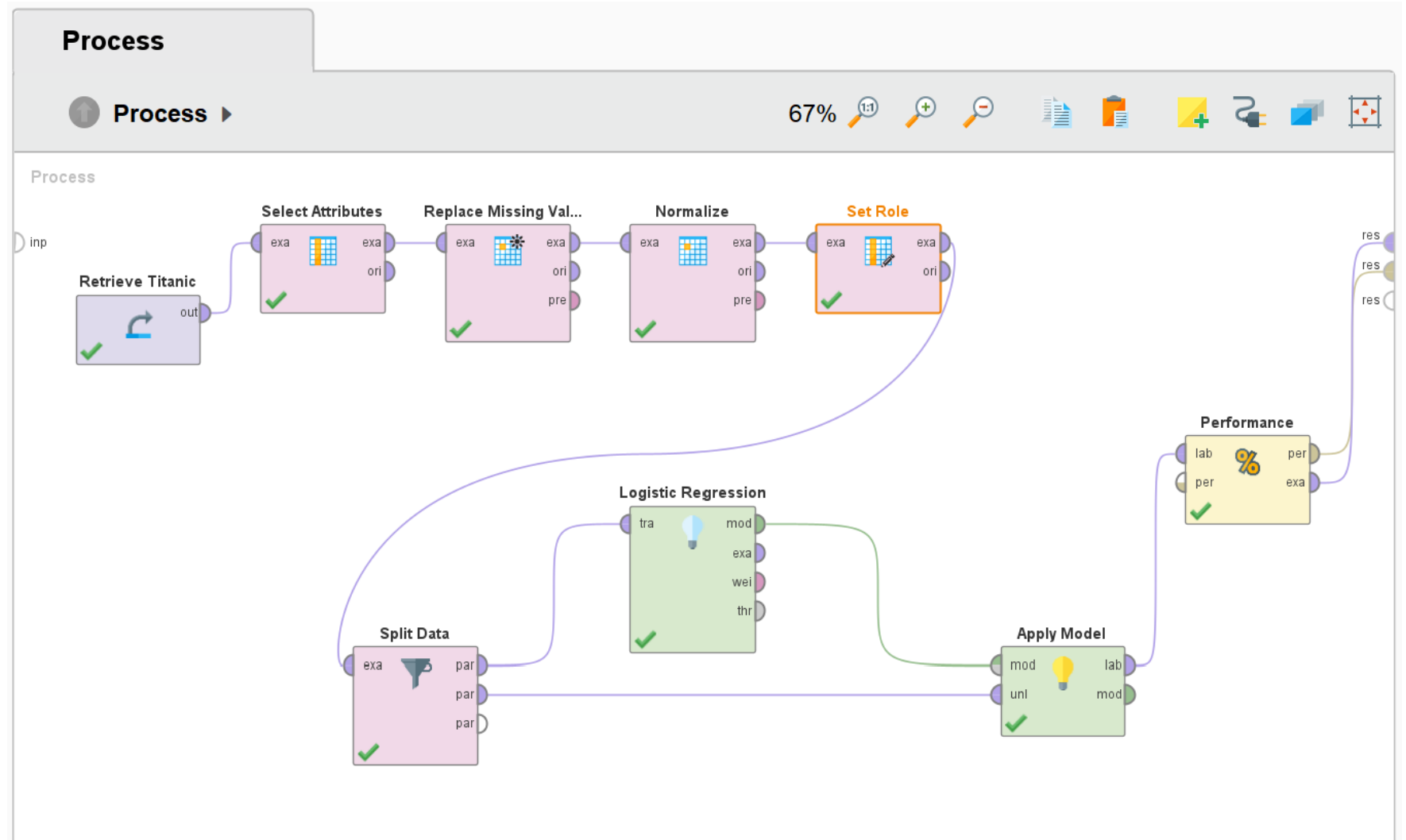
2602624891

Tugas

1. Klasifikasi dengan 1 metode untuk titanic dengan menggunakan Rapid Miner dan Python. Bandingkan hasilnya. Tunjukkan performancinya
2. Klasifikasi dengan 1 metode untuk iris dengan menggunakan Rapid Miner dan Python. Bandingkan hasilnya. Tunjukkan performancinya. Pahami multi class dan binary class
3. <https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>. Lakukan feature selection dengan PCA dengan 1 metode klasifikasi. Rapid Miner dan program

Titanic Dataset

- Design



Titanic Dataset

- Design

Process

Edit Parameter List: set roles

Edit Parameter List: **set roles**
This parameter defines new attribute roles.

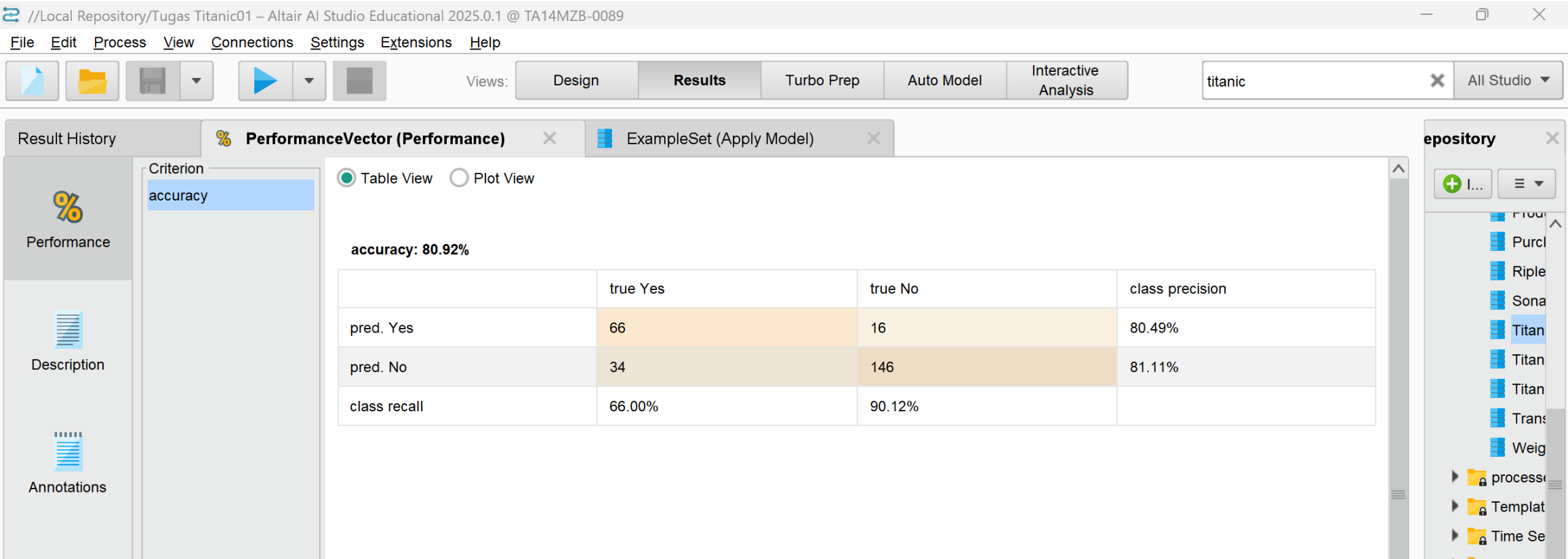
attribute name	target role
Survived	label
Age	regular
Sex	regular

 Add Entry  Remove Entry  Apply  Cancel

Leverage the Wisdom of Crowds to get operator recommendations based on your process design!

Titanic Dataset

- Result



Titanic Dataset

- Perbandingan dengan Python

accuracy: 80.92%

	true Yes	true No	class precision
pred. Yes	66	16	80.49%
pred. No	34	146	81.11%
class recall	66.00%	90.12%	

... Accuracy: 0.9580

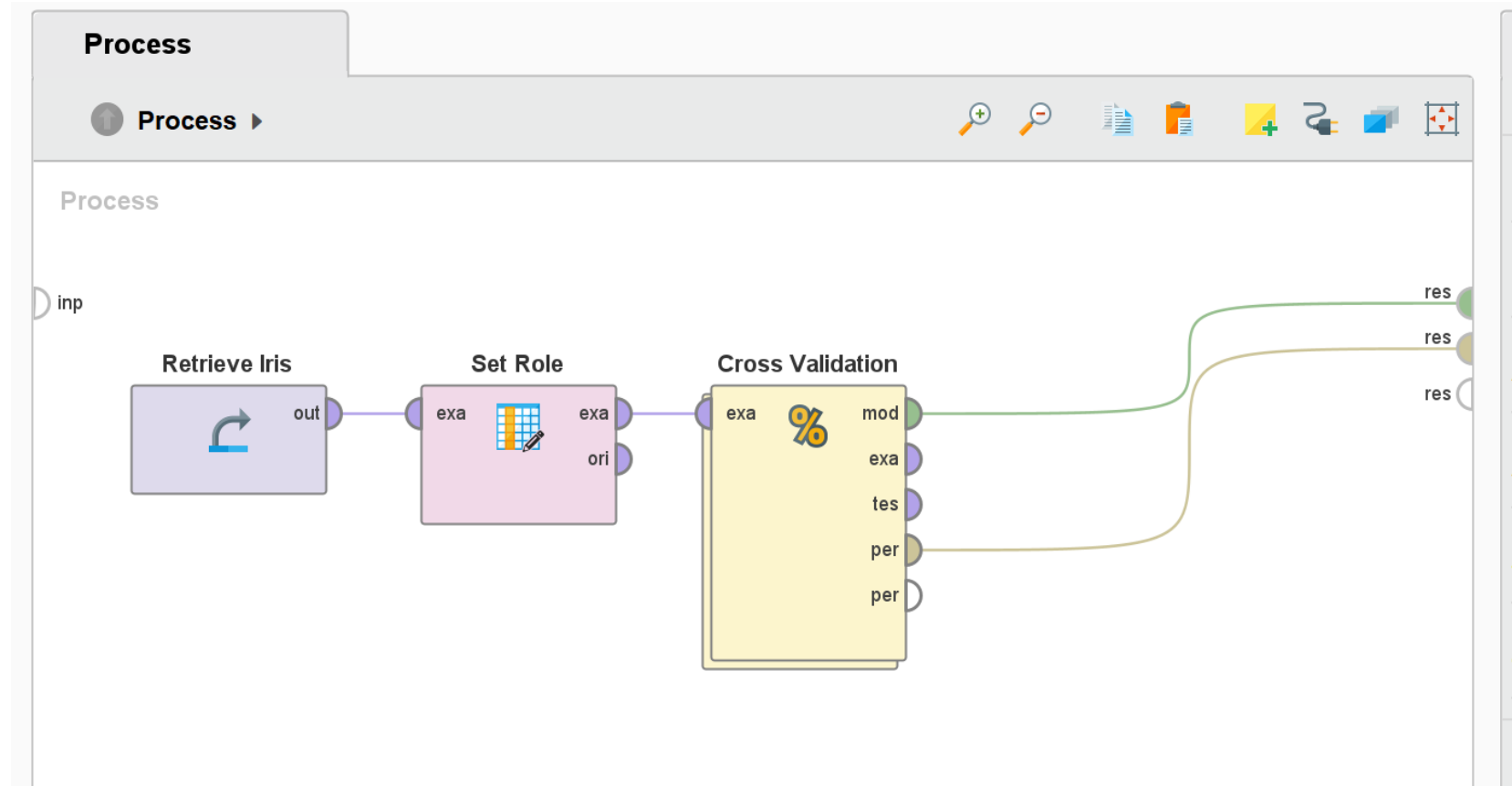
Classification Report:

	precision	recall	f1-score	support
0	0.93	1.00	0.96	144
1	1.00	0.91	0.95	118
accuracy			0.96	262
macro avg	0.96	0.95	0.96	262
weighted avg	0.96	0.96	0.96	262

Hasil yang diraih ketika menggunakan python rata-rata lebih tinggi

Iris Dataset

- Design



Iris Dataset

- Design

Proc

↑

Proc

inp

 Edit Parameter List: set roles ✕

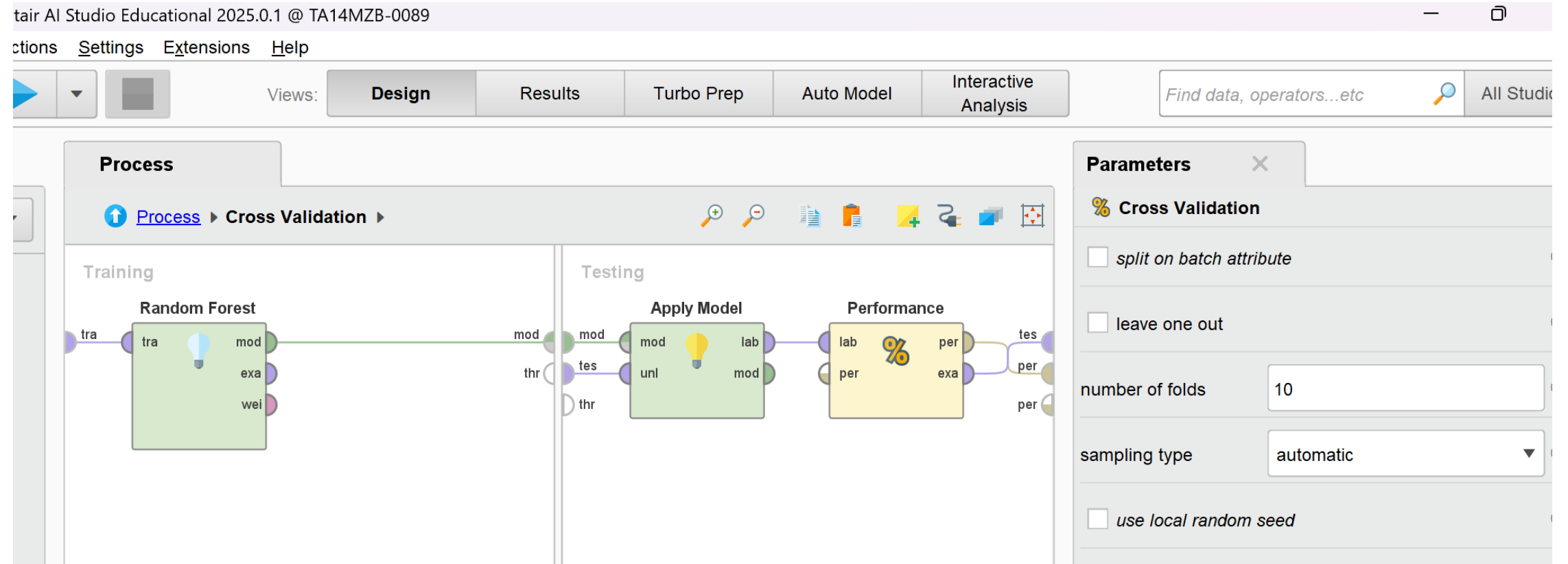


Edit Parameter List: **set roles**
This parameter defines new attribute roles.

attribute name	target role
id ▼	regular ▼
label ▼	label ▼

Iris Dataset

- Design



Iris Dataset

- Result

Result History

%

Performance

Description

Annotations

Criterion

accuracy

PerformanceVector (Performance)

Random Forest Model (Random Forest)

☒ Table View

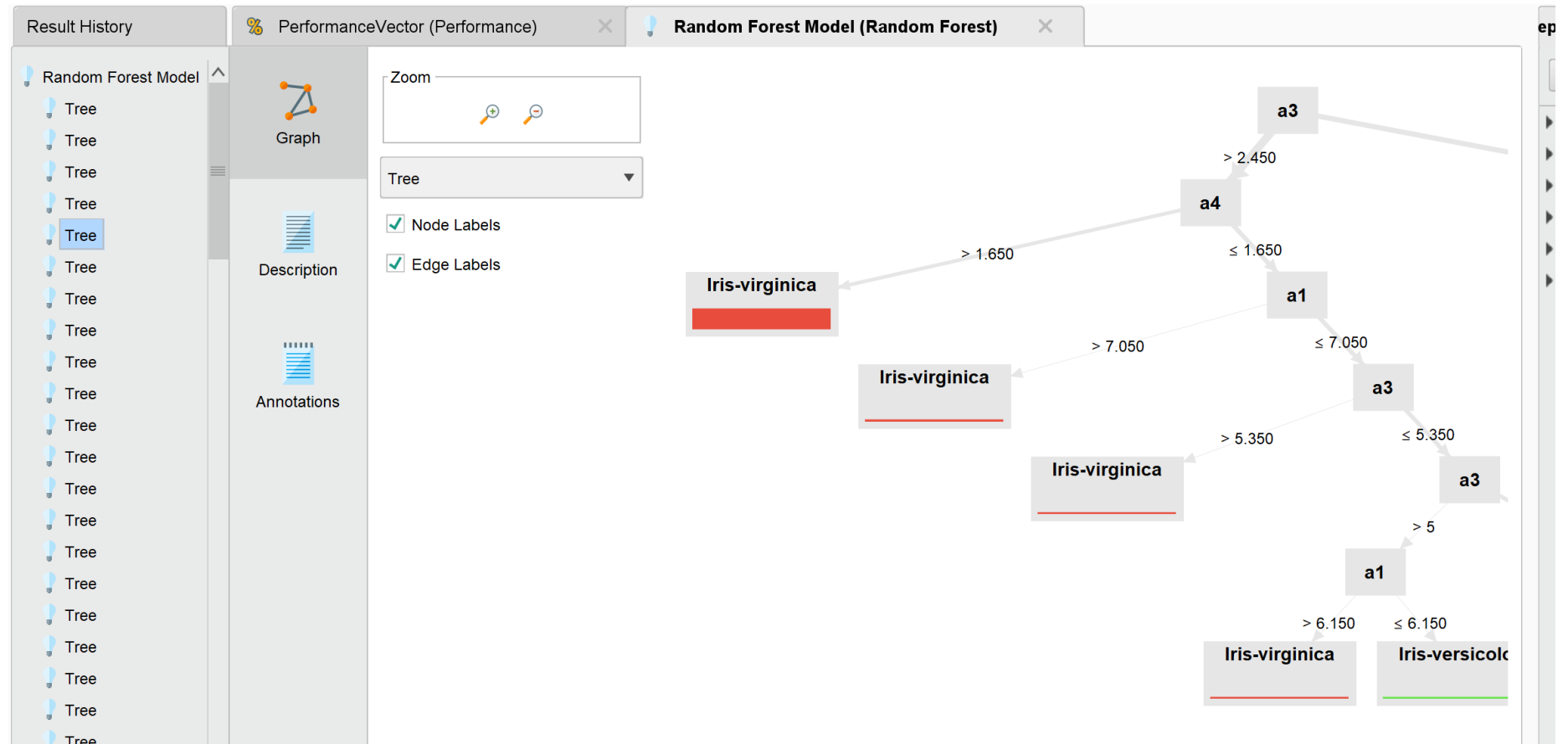
☐ Plot View

accuracy: 96.00% +/- 4.66% (micro average: 96.00%)

	true Iris-setosa	true Iris-versicolor	true Iris-virginica	class precision
pred. Iris-setosa	50	0	0	100.00%
pred. Iris-versicolor pred. Iris-setosa	0	47	3	94.00%
pred. Iris-virginica	0	3	47	94.00%
class recall	100.00%	94.00%	94.00%	

Iris Dataset

- Result



Iris Dataset

- Perbandingan dengan Python

Model: RandomForestClassifier

Accuracy: 0.9556

Classification Report:

	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	16
Iris-versicolor	0.91	0.91	0.91	11
Iris-virginica	0.94	0.94	0.94	18
accuracy			0.96	45
macro avg	0.95	0.95	0.95	45
weighted avg	0.96	0.96	0.96	45

Model: SVC

Accuracy: 0.9778

Classification Report:

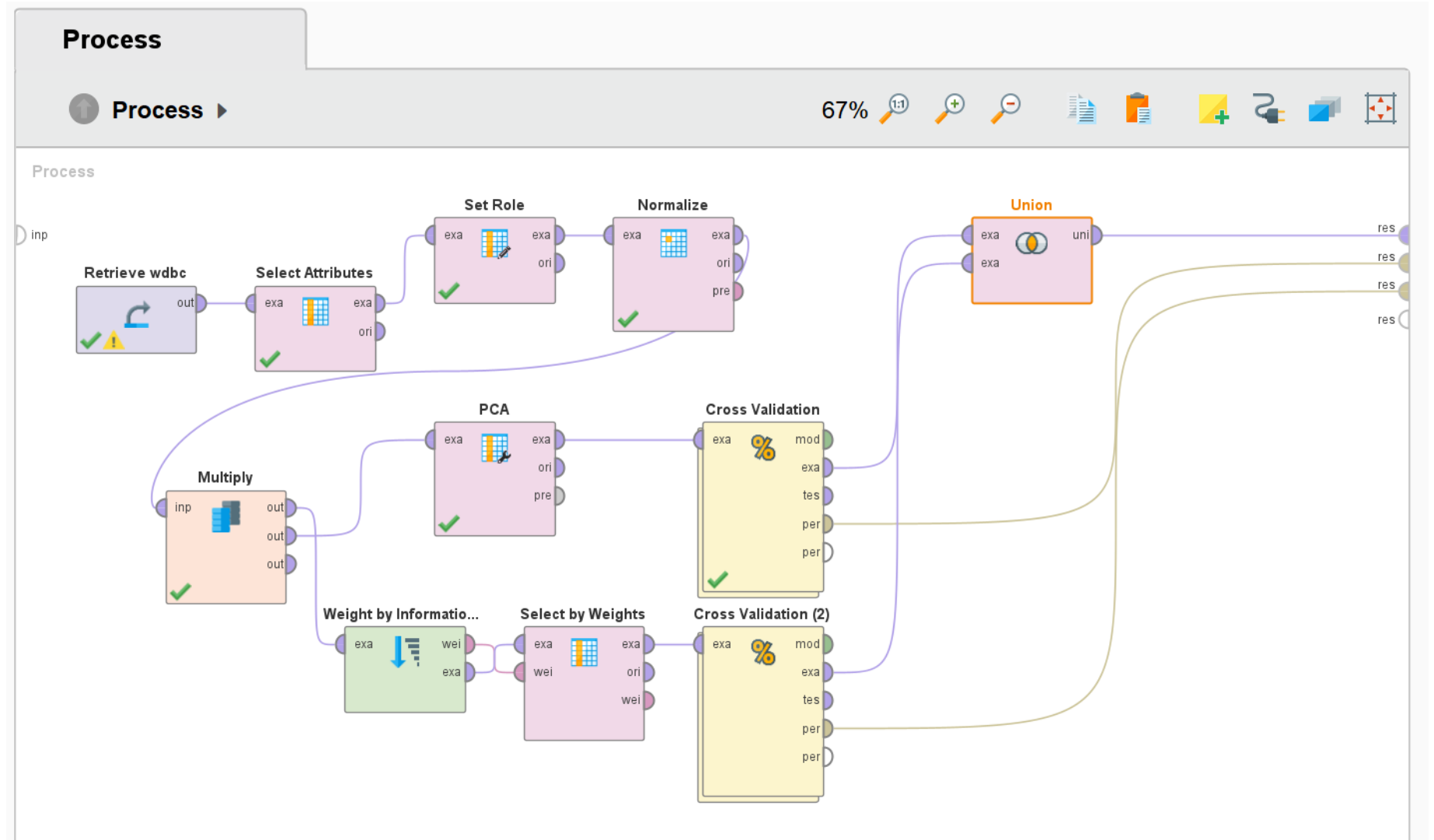
	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	16
Iris-versicolor	0.92	1.00	0.96	11
Iris-virginica	1.00	0.94	0.97	18
accuracy			0.98	45
macro avg	0.97	0.98	0.98	45
weighted avg	0.98	0.98	0.98	45

PerformanceVector (Performance)			Random Forest Model (Random Forest)	
			☒ Table View	☐ Plot View
			accuracy: 96.00% +/- 4.66% (micro average: 96.00%)	
	true Iris-setosa	true Iris-versicolor		
pred. Iris-setosa	50	0		
...	...	...		

Ketika menggunakan rapidminer, didapatkan hasil akurasi 96%, sementara Ketika menggunakan Python, menggunakan model yang sama yaitu random forest didapatkan hasil yang lebih rendah, tetapi didapatkan hasil akurasi 97% Ketika menggunakan model SVC / Support Vector Machine Classifier

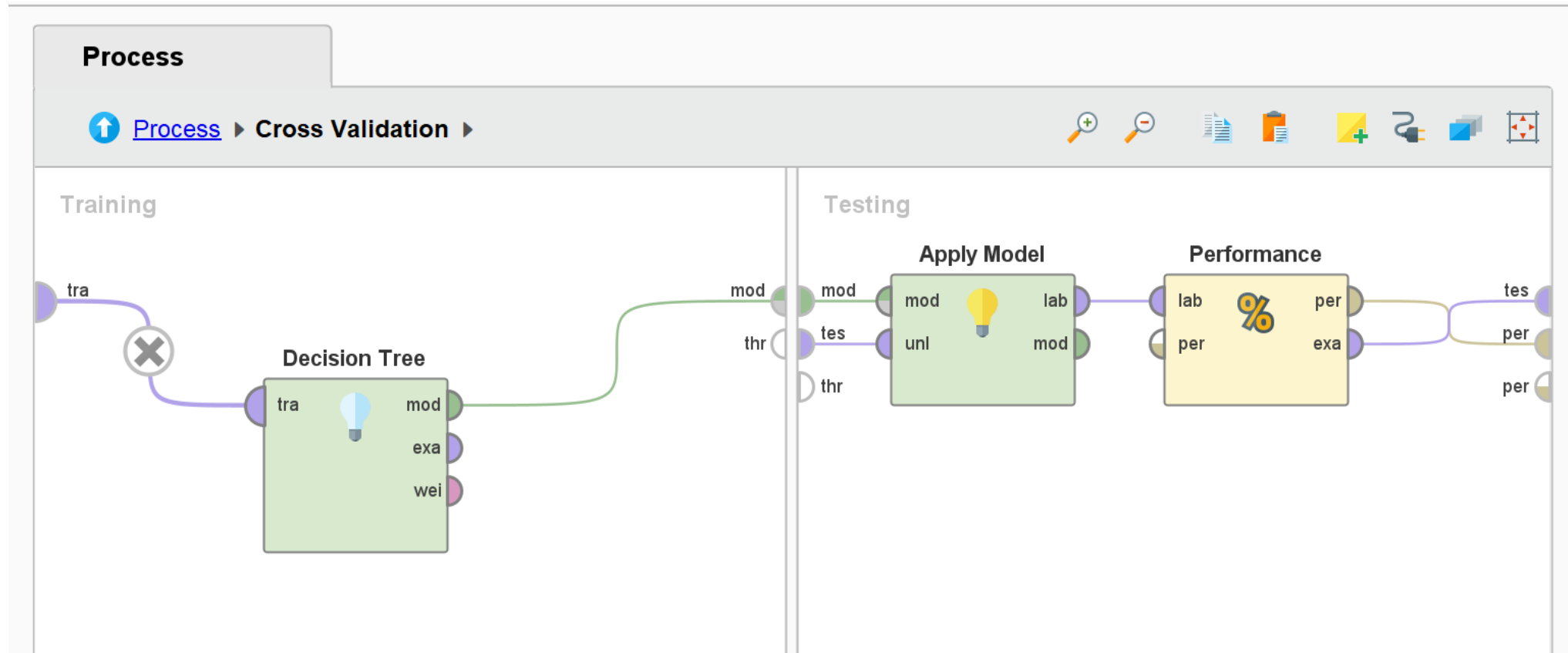
Wisconsin Breast Cancer Dataset

- Design



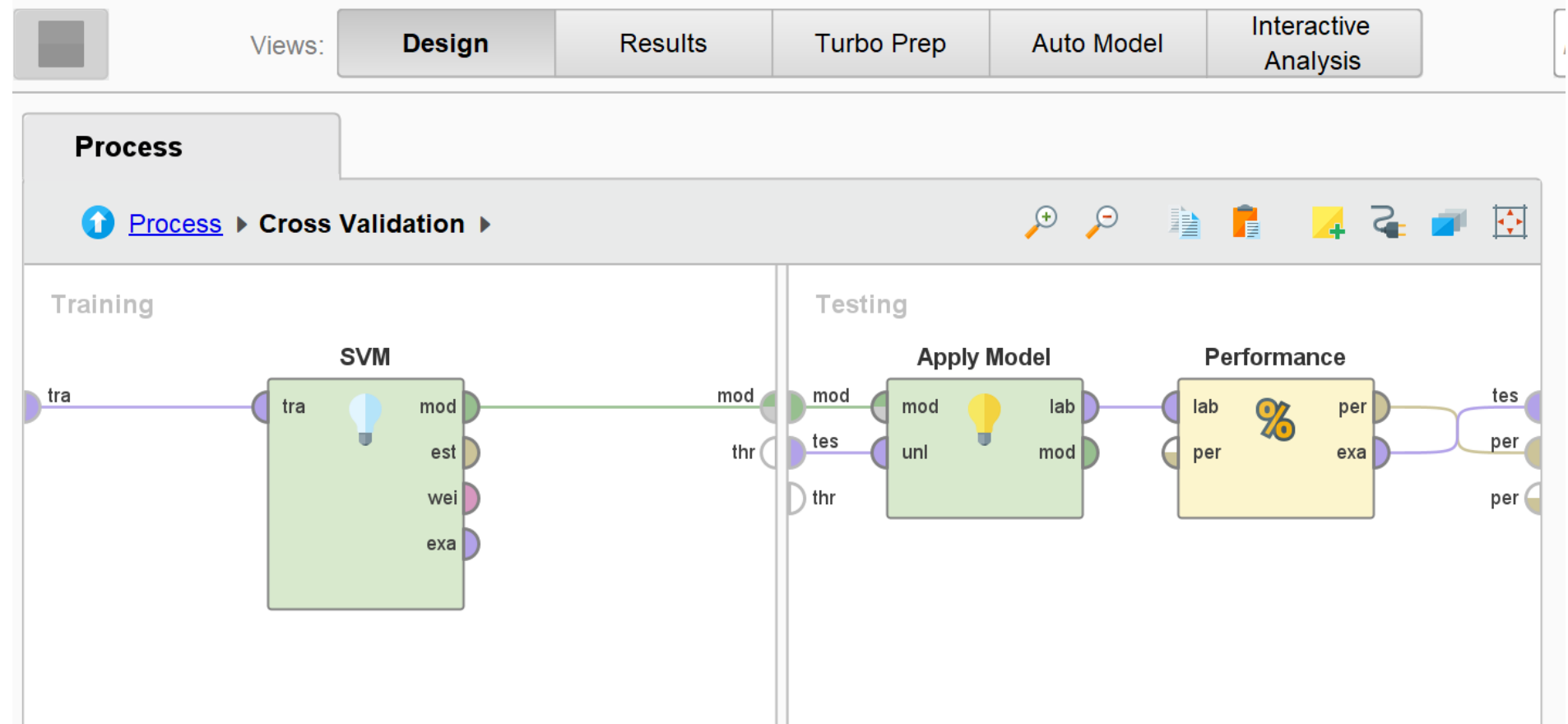
Wisconsin Breast Cancer Dataset

- Design



Wisconsin Breast Cancer Dataset

- Design



Wisconsin Breast Cancer Dataset

- Result (Principal Component Analysis)

Result History

%

Performance

Document Icon

Description

Annotations Icon

Annotations

Criterion

accuracy

PerformanceVector (Performance (2))

PerformanceVector (Performance)

ExampleSet (Union)

☒ Table View ☐ Plot View

accuracy: 97.01% +/- 2.49% (micro average: 97.01%)

	true M	true B	class precision
pred. M	196	2	98.99%
pred. B	15	355	95.95%
class recall	92.89%	99.44%	

	true M	true B	class precision
pred. M	196	2	98.99%
pred. B	15	355	95.95%
class recall	92.89%	99.44%	

Wisconsin Breast Cancer Dataset

- Result (Select by Weight)

Result History

%

Performance

Description

Annotations

Criterion

accuracy

% PerformanceVector (Performance (2))

% PerformanceVector (Performance)

ExampleSet (Union)

☒ Table View

☐ Plot View

accuracy: 90.14% +/- 2.77% (micro average: 90.14%)

	true M	true B	class precision
pred. M	170	15	91.89%
pred. B	41	342	89.30%
class recall	80.57%	95.80%	

Wisconsin Breast Cancer Dataset

- Result

Result History PerformanceVector (Performance (2)) PerformanceVector (Performance) ExampleSet (Union)

Open in Turbo Prep Auto Model Interactive Analysis Filter (1,136 / 1,136 examples): all

Row No.	Diagnosis	pc_1	pc_2	pc_3	pc_4	pc_5	pc_6	pc_7	pc_8
1	M	2.410	-3.746	-0.583	-1.155	0.595	-0.146	0.282	-0.072
2	M	5.772	-1.055	-0.539	-0.941	-0.217	-0.440	-0.470	0.672
3	M	7.107	10.323	-3.143	-0.120	-2.995	-2.594	-1.937	-1.388
4	M	3.968	-1.956	1.420	-2.979	0.473	1.180	0.185	0.936
5	M	2.370	3.987	-2.917	-0.968	-1.097	0.466	0.022	-0.529
6	M	2.262	-2.660	-1.686	-0.183	0.029	0.046	0.258	0.266
7	M	2.164	2.345	-0.806	0.135	-1.447	1.366	-0.133	-0.967
8	M	3.178	3.442	-3.127	0.582	-1.533	-0.522	-0.170	0.178
9	M	6.367	7.786	-4.284	3.287	1.749	1.076	-0.798	-2.431
10	M	-0.785	-2.646	-0.539	1.668	0.315	-0.178	0.161	-0.635
11	M	2.678	0.076	-1.494	-0.060	0.319	-0.530	-0.856	-0.604
12	M	8.208	2.736	5.634	1.081	1.093	-2.887	0.581	-1.178
13	M	0.365	-0.960	1.660	0.574	0.484	-1.102	0.019	1.811
14	M	4.363	4.898	-2.768	1.385	1.295	0.581	-0.516	-0.709
15	M	4.104	3.013	-3.085	2.437	-0.387	-0.183	-1.003	-0.831

ExampleSet (1,136 examples,1 special attribute,12 regular attributes)

repository Training Resources Samples Local Repository Temporary Repository Community Samples DB (Legacy)

Wisconsin Breast Cancer Dataset

- Perbandingan dengan Python

```
SVM Accuracy: 0.9766
Classification Report:
              precision    recall  f1-score   support

         B            0.97         0.99         0.98         103
         M            0.98         0.96         0.97          68

 accuracy              0.98              0.98         171
 macro avg              0.98              0.97         171
 weighted avg              0.98              0.98         171
```

PerformanceVector (Performance (2))

PerformanceVector (Performance)

☒ Table View ☐ Plot View

accuracy: 90.14% +/- 2.77% (micro average: 90.14%)

	true M	true B
pred. M	170	15
pred. B	41	342
class recall	80.57%	95.80%

PerformanceVector (Performance (2))

PerformanceVector (Performance)

ExampleSet (Union)

☒ Table View ☐ Plot View

accuracy: 97.01% +/- 2.49% (micro average: 97.01%)

	true M	true B	class precision
pred. M	196	2	98.99%
pred. B	15	355	95.95%
class recall	92.89%	99.44%	

Akurasi SVM menggunakan Python dan Rapidminer relatif sama, tetapi terdapat penurunan performa Ketika menggunakan feature selection menjadi 90% akurasi.